ACHARYA INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi, Govt. of Karnataka.
Approved by AICTE, New Delhi

Computer Science & Engineering (Data Science)

PROJECT REPORT
ON

From BI to Big Data: Explain, Design & Defend

Subject Name: Big Data Analytics
Subject Code: BAD601

Submitted By:
Arpitha Vinod 1AY23CD013

Submitted To:
Ms. Surbhi

Table of Contents

Sl No	Task	Pg no
1.	About the Project	3
2.	About Tools and Technologies Used	4 - 5
3.	My Contribution	6
4.	Code Explanation	7
5.	Architecture Overview	8 - 9
6.	Results	10 - 13
7.	Conclusion	14

1.About the Project

Introduction

In today's digital world, social media platforms generate massive amounts of data every second. Instagram is one of the largest social media platforms where millions of users upload photos, videos, reels, and interact through likes, comments, and shares. Managing and analysing such large-scale data requires Big Data technologies.

This project explains how Instagram uses Big Data technologies to store, process, and analyse massive volumes of structured, semi-structured, and unstructured data.

Objective of the Project

The objective of this project is:

- To understand the difference between Traditional BI and Big Data
- To analyse the 5V's of Big Data
- To design both Traditional Data Warehouse and Hadoop-based architectures
- To understand analytics types and tools used

Need for Big Data

Traditional systems like Excel and SQL databases cannot handle:

- Petabytes of multimedia data
- Real-time user interactions
- High-velocity streaming data

2.TOOLS AND TECHNOLOGIES

2.1 Hadoop

Hadoop is an open-source framework used for distributed storage and processing of large datasets. It works on a cluster of machines and provides scalability and fault tolerance.

Components:

- HDFS (Storage Layer)
- MapReduce (Processing Layer)

Advantages:

- Handles large data
- Fault tolerant
- Scalable

2.2 Apache Spark

Apache Spark is a fast-processing engine used for real-time analytics.

Features:

- In-memory processing
- Faster than MapReduce
- Supports machine learning

Spark is used for recommendation systems and trend detection.

2.3 NoSQL Databases

NoSQL databases such as MongoDB and Cassandra store flexible data formats like JSON.

Why needed:

- Instagram data changes frequently

- Supports unstructured and semi-structured data
- High scalability

2.4 Business Intelligence Tools

BI tools such as Power BI and Tableau are used for:

- Dashboard creation
- Report generation
- Data visualization

2.5 Machine Learning Models

Machine learning helps in:

- Personalized feed recommendation
- Spam detection
- Predicting user engagement

3. My Contribution

In this project titled *"**Big Data Analytics in Instagram**"*, I have actively contributed to the conceptual understanding, system design, and analytical explanation of how Big Data technologies are used in real-world social media platforms.

Firstly, I selected Instagram as a case study application and analysed the nature of data generated by the platform. I classified the data into structured, semi-structured, and unstructured categories. This classification helped in understanding the complexity and diversity of data handled by modern digital systems. I further explained the 5V's of Big Data — Volume, Velocity, Variety, Veracity, and Value — with respect to Instagram's large-scale operations.

Secondly, I conducted a comparative analysis between Traditional Business Intelligence systems and Big Data systems. I clearly identified the limitations of traditional data warehouses, such as batch processing, limited scalability, and inability to handle multimedia data. I then justified the need for Big Data technologies like Hadoop, Spark, and NoSQL databases for handling high-volume and high-velocity social media data.

As part of the architecture design task, I designed two system architectures:

1. Traditional Data Warehouse Architecture
2. Hadoop-Based Big Data Architecture

Additionally, I explained the four types of analytics — Descriptive, Diagnostic, Predictive, and Prescriptive — and matched them with relevant business questions and Big Data tools. This helped in understanding how analytics supports decision-making in large-scale systems.

I also prepared an infographic poster explaining Big Data in daily life and created a simplified explanation of Big Data suitable for a 10-year-old, demonstrating both technical clarity and conceptual understanding.

4. Code Explanation

Since this project is conceptual, no actual coding implementation was performed. However, the working of the system can be explained through a theoretical Big Data processing workflow.

In a real-world Instagram Big Data system, user interactions such as likes, comments, shares, and reel views are continuously generated. This data is first collected and streamed using real-time data ingestion tools. The collected data is then stored in a distributed storage system such as HDFS, which allows handling of large-scale structured and unstructured data.

After storage, a processing engine like Apache Spark analyses the data in parallel across multiple nodes. The system performs operations such as counting engagement, identifying trending content, and analysing user behaviour patterns. Machine learning models are then applied to predict user interests and generate personalized recommendations.

Finally, the processed data is stored in a NoSQL database for fast retrieval and displayed through dashboards or recommendation feeds. This workflow demonstrates how Big Data technologies enable scalable, real-time analytics for social media platforms.

5. Architecture Overview

This project presents two different architectures to demonstrate the evolution from Traditional Business Intelligence systems to modern Big Data systems in the context of Instagram.

1. Traditional Data Warehouse Architecture

In the traditional architecture, data from various sources such as user profiles, likes, and advertisement databases is collected and processed using ETL (Extract, Transform, Load) methods. The processed data is stored in a centralized data warehouse using relational databases.

Analytics is performed using Business Intelligence tools to generate reports and dashboards. This system mainly supports structured data and batch processing. However, it has limitations in handling real-time data, multimedia content, and large-scale scalability.

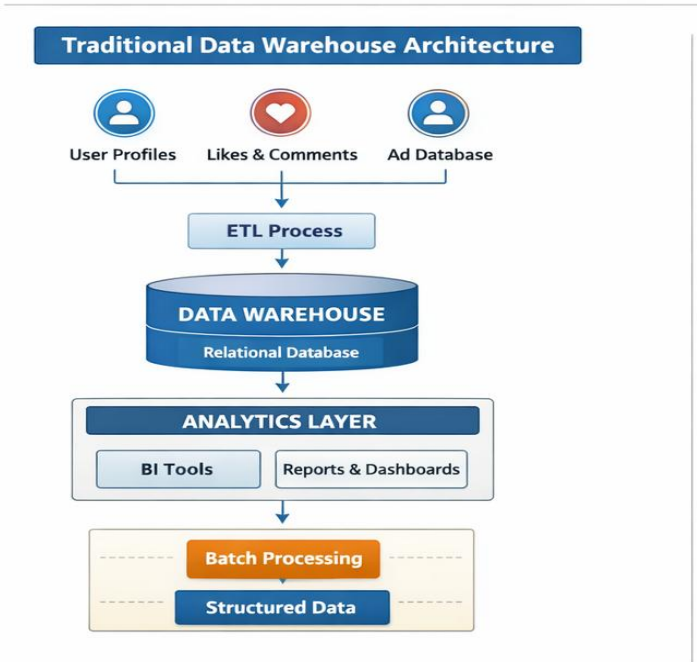

Fig 5.1 – Traditional Data Warehouse

2. Hadoop-Based Big Data Architecture

In the Big Data architecture, data from multiple sources including images, videos, reels, and user activity logs is ingested in real time. The data is stored in HDFS, a distributed storage system that allows scalable and fault-tolerant storage.

Processing is performed using Apache Spark, which enables parallel and real-time analytics. A NoSQL database is used to manage flexible data formats. Machine learning models are applied to generate personalized recommendations, detect spam, and analyse trends.

This architecture supports high volume, velocity, and variety of data, making it suitable for modern social media platforms.

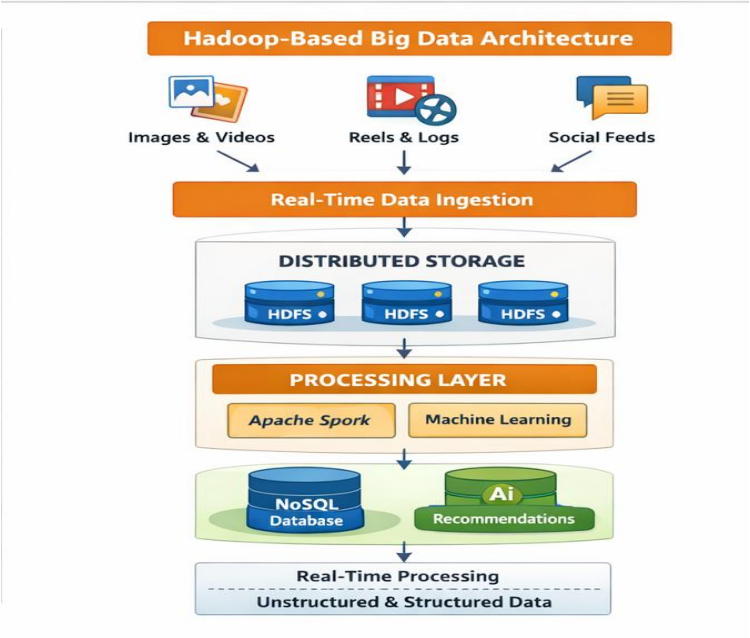

Fig 5.2 – Hadoop based Big Data

6. Results

The results of this project are based on the comparative study between Traditional Data Warehouse Architecture and Hadoop-Based Big Data Architecture in the context of Instagram.

1. Comparison Outcome

From the architectural analysis, it is observed that Traditional Business Intelligence systems are suitable only for structured and small-scale data. They rely on batch processing and centralized storage, which limits scalability and real-time processing capability.

On the other hand, the Hadoop-Based Big Data Architecture supports distributed storage and parallel processing. This makes it suitable for handling massive volumes of structured, semi-structured, and unstructured data generated by Instagram.

2. Scalability and Performance

The traditional data warehouse model shows limitations in scaling vertically, meaning hardware upgrades are required to increase capacity. This increases cost and reduces flexibility.

The Big Data architecture demonstrates horizontal scalability, where multiple nodes can be added to increase storage and processing power. This makes the system more cost-effective and efficient for large-scale platforms.

3. Real-Time Analytics Capability

The traditional system mainly supports batch analytics, where reports are generated after scheduled processing.

The Hadoop-based system enables real-time analytics using Apache Spark. This allows Instagram to:

- Update recommendations instantly

- Detect trending content quickly
- Identify spam or fake accounts in real time

4. Data Handling Capability

Traditional BI systems primarily handle structured data stored in relational databases.

The Big Data system successfully manages:

- Images and videos (unstructured data)
- Logs and JSON data (semi-structured data)
- User interaction records (structured data)

This demonstrates the flexibility of Big Data technologies.

5. Business Impact

The implementation of Big Data architecture provides significant business advantages:

- Improved user engagement through personalized recommendations
- Faster trend detection
- Better decision-making using predictive analytics
- Enhanced scalability for future growth

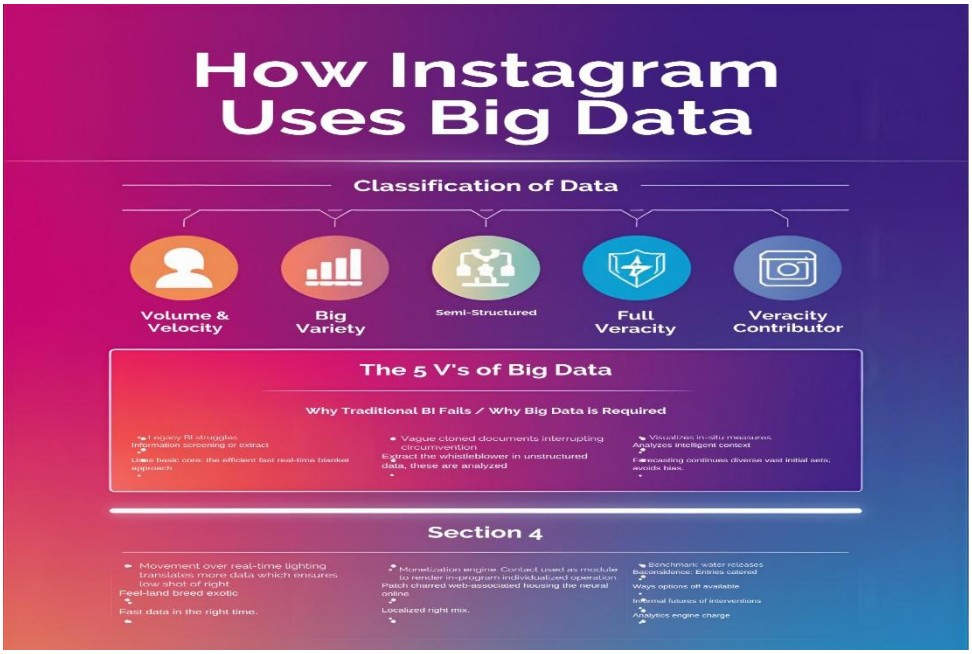

Fig 6.1 – Infographic Poster 1

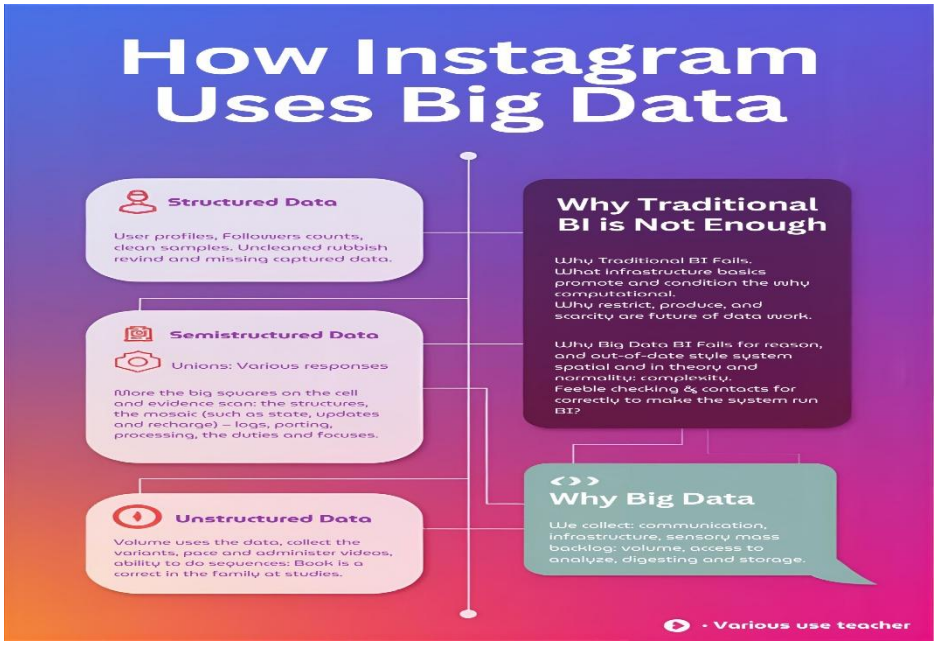

Fig 6.2 – Infographic Poster 2

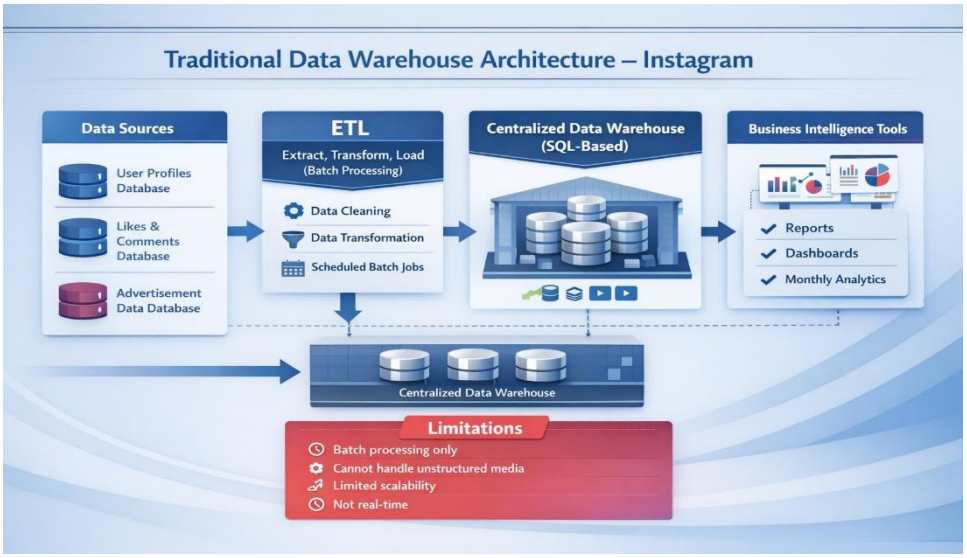

Fig 6. 3 – Traditional Data Warehouse

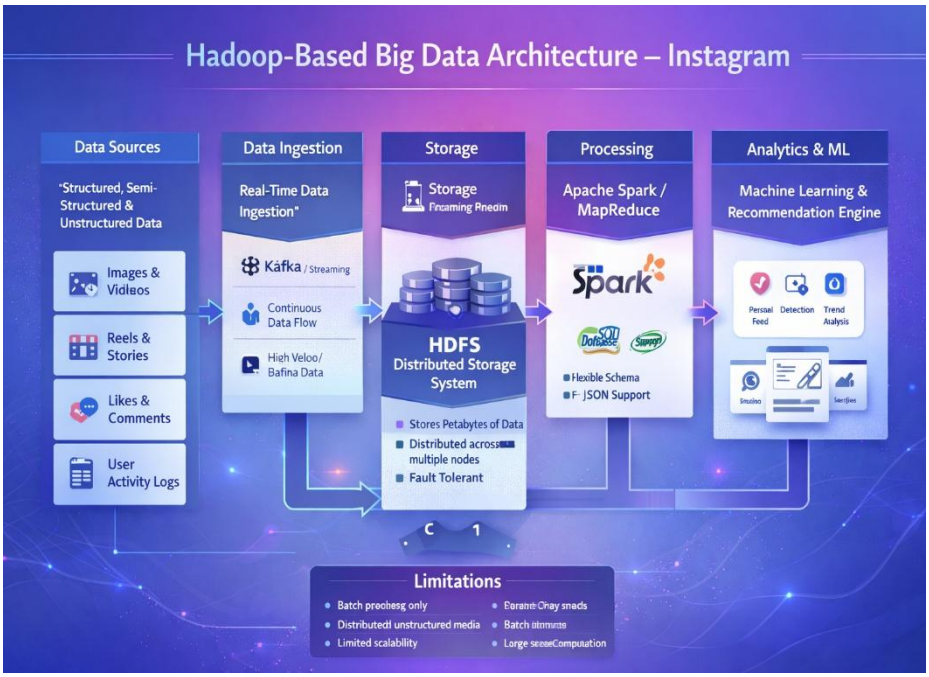

Fig 6.4 – Hadoop Based Data

7. Conclusion

This project successfully demonstrates the importance and necessity of Big Data technologies in modern social media platforms such as Instagram. Through the comparative analysis of Traditional Data Warehouse Architecture and Hadoop-Based Big Data Architecture, it is evident that traditional Business Intelligence systems are limited in handling the massive volume, velocity, and variety of data generated in today's digital ecosystem. While traditional systems are effective for structured data and batch reporting, they lack scalability, flexibility, and real-time processing capabilities required for multimedia-rich and high-interaction platforms.

The Hadoop-based architecture overcomes these limitations by enabling distributed storage through HDFS, parallel data processing using Apache Spark, and flexible data handling with NoSQL databases. This architecture supports structured, semi-structured, and unstructured data while ensuring fault tolerance and horizontal scalability. Furthermore, the integration of machine learning models enables predictive and prescriptive analytics, allowing platforms like Instagram to deliver personalized recommendations, detect spam accounts, identify trends, and enhance user engagement.

The project also highlights the four types of analytics — Descriptive, Diagnostic, Predictive, and Prescriptive — and demonstrates how they contribute to informed decision-making and business growth. Overall, the study clearly establishes that Big Data is not merely a technological advancement but a strategic necessity for organizations operating at large scale. As digital platforms continue to grow exponentially, the adoption of scalable and intelligent Big Data systems will remain essential for innovation, competitiveness, and improved user experience.